FORMAL SMOOTHNESS OF AN EXTENSION OF DISCRETE VALUATION RINGS IS INHERITED BY FINITE EXTENSIONS OF THE FRACTION FIELD

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Editorial note. The counted statement span carries a display of the base changed residue ring in the proof and a threefold indexing notation (m_ij); both are delivered verbatim. The two references inside the counted statement and proof spans are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The equivalence characterizing formal smoothness of an extension of discrete valuation rings in terms of the separable residue field extension. invoked at proof lines 34016-34017. Is the criterion that must be verified for each of the base changed extensions.
- The construction remark describing the extensions (A_1)_{m_i} in (B_1)_{m_ij}. referred to at statement lines 34011-34012. Fixes the rings whose formal smoothness is asserted.
- Separability of the residue field extension kappa_A in kappa_B. used at proof lines 34019-34024. Makes the residue ring a product of separable field extensions, giving the nonzerodivisor property of pi_i.

Source reference. Original lemma 32989: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32989–32989. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. Assume that $A \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology. Then for any finite extension K_1/K we have $L_1 = L \otimes_K K_1$, $B_1 = B \otimes_A A_1$, and*

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each extension $(A_1)_{\mathfrak{m}_i} \subset (B_1)_{\mathfrak{m}_{ij}}$ *(see Remark 33882) is formally smooth in the* $\mathfrak{m}_{ij}$*-adic topology.*

Proof. We will use the equivalence of Lemma 32989 without further mention. Let $\pi \in A$ and $\pi_i \in (A_1)_{\mathfrak{m}_i}$ be uniformizers. As $\kappa_A \subset \kappa_B$ is separable, the ring

$$(B \otimes_A (A_1)_{\mathfrak{m}_i})/\pi_i(B \otimes_A (A_1)_{\mathfrak{m}_i}) = B/\pi B \otimes_{A/\pi A} (A_1)_{\mathfrak{m}_i}/\pi_i(A_1)_{\mathfrak{m}_i}$$

is a product of fields each separable over $\kappa_{\mathfrak{m}_i}$. Hence the element π_i in $B \otimes_A (A_1)_{\mathfrak{m}_i}$ is a nonzerodivisor and the quotient by this element is a product of fields. It follows that $B \otimes_A A_1$ is a Dedekind domain in particular reduced. Thus $B \otimes_A A_1 \subset B_1$ is an equality. $\square$